

ERIC (CHEN-CHIEN) CHANG

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SUMMARY

- Backend / Data Infrastructure Engineer specializing in **performance tuning**, **distributed systems**, and **open-source data platforms**.
- Core stack: **Go**, **Java**, **Rust**, **TypeScript**, **PostgreSQL**, **Kafka**, **Docker**, **Kubernetes**. Delivered measurable wins including first-byte latency **100s to 5s**, WebUI load time **60s to 1s**, and simulation jobs **2hr to 1min**.

OPEN SOURCE CONTRIBUTIONS

- **Apache Gravitino Committer** ([GitHub](#)): implemented PostgreSQL/MySQL container-suite management for integration tests, built reproducible Docker image workflows, reduced Hive image size by **420MB**, and ported partition APIs to the Python client.
- **Apache Kafka Contributor** ([GitHub](#)): fixed JVM container awareness in producer perf tests and corrected partition switching so offsets are committed before partition changes: [PR #21584](#).
- **Apache DataFusion and Comet Contributor**: implemented Spark str-to-map in Rust ([PR #20120](#)) and wired Comet's native SparkStrToMap UDF with SQL file coverage ([PR #3654](#)).
- **redis/rueidis Contributor** ([PR #395](#)): implemented the go-redis API Adapter for teams migrating from go-redis.

EXPERIENCE

Lawson

January 2024 - December 2024

Backend Software Engineer. Tech Stack: **TypeScript**, **NodeJS**, **Docker**, **MongoDB**, **Prometheus**, **Grafana** Taipei, Taiwan

- Implemented **HTTP streaming** for CPU-intensive computation results, reducing first-byte latency from **100s** to **5s**.
- Built a **MinIO events** and **Elasticsearch** monitoring path to detect missing data proactively and improve pipeline reliability.
- Profiled and fixed a **NAPI-rs** performance issue with **Pyroscope**, improving Node.js native package execution.
- Built an **LLM service** for extracting information from ambiguous regulatory text and automated government act crawling, re-crawl, and re-parse workflows.

Mediatek (Contract)

February 2022 - March 2023

Backend Software Engineer. Tech Stack: **Go**, **Linux**, **Docker**, **PostgreSQL** Hsinchu, Taiwan

- Maintained a low-power simulation and benchmarking data warehouse, improving WebUI page load time from **60s** to **1s** with **PostgreSQL index-only scans**: [Blog Post](#).
- Used **Pyroscope eBPF** to identify import bottlenecks, then applied **PostgreSQL bulk import** to reduce simulation job duration from **2hr** to **1min**.
- Designed a message-queue-based microservice architecture and operated a **5-node** service cluster for horizontally scalable long-running jobs.

- Reduced Go service memory consumption by **50%** using **sync.Pool**: [Blog Post](#).
- Built integration-test fixture factories and maintained observability with **Prometheus**, **Grafana**, **OpenTelemetry**, and **Pyroscope**.

National Chen-Kung University

September 2017 - July 2019

*Research Assistant. Tech Stack: **Python**, **C***

Tainan, Taiwan

- Conducted FDM 3D printing research across firmware, filament experiments, and hardware/software integration.
- Built experimental tooling with **Python**, **C**, **MicroPython**, and **Arduino**.

EDUCATION

National Taiwan Ocean University, Electrical Engineering coursework (Drop Out)

2013 - 2017

SKILLS

Languages: Go, Java, Rust, TypeScript, Python, C

Data Warehouse / Lakehouse: PostgreSQL, ClickHouse, Hive, StarRocks

Streaming / Processing: Kafka, Kafka Streams, Flink, Spark, SQL, dbt

Runtime / Orchestration: Airflow, Docker, Kubernetes, Linux

Observability / Data Quality: Prometheus, Grafana, OpenTelemetry, Pyroscope, Great Expectations

DevOps / Applied AI: Git, GitHub Actions, LLM service integration